

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285905

Kind Code

A1

Publication Date

September 11, 2025

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SEMICONDUCTOR PACKAGE AND MANUFACTURING METHOD THEREOF

Abstract

A semiconductor package includes a semiconductor device, an encapsulating material encapsulating the semiconductor device, and a redistribution structure disposed over the encapsulating material and the semiconductor device. The semiconductor device includes conductive bumps and a dielectric film encapsulating the conductive bumps, where a material of the dielectric film comprises an epoxy resin and a filler. The conductive bumps are isolated from the encapsulating material by the dielectric film. The redistribution structure is electrically connected to the conductive bumps.

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Appl. No.: 19/215281

Filed: May 21, 2025

Related U.S. Application Data

parent US continuation 18356212 20230720 parent-grant-document US 12322640 child US 19215281

parent US continuation 17525975 20211115 parent-grant-document US 11776838 child US 18356212

parent US division 16547605 20190822 parent-grant-document US 11177156 child US 17525975

Publication Classification

Int. Cl.: H01L21/683 (20060101); **H01L21/56** (20060101); **H01L23/00** (20060101); **H10D84/01** (20250101)

U.S. Cl.:

CPC H01L21/6836 (20130101); **H01L21/568** (20130101); **H01L24/05** (20130101); **H01L24/13** (20130101); **H10D84/01** (20250101); H01L2221/6834 (20130101); H01L2221/68377 (20130101); H01L2221/68386 (20130101); H01L2224/0231 (20130101); H01L2224/02381 (20130101); H01L2224/10122 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 18/356,212, filed on Jul. 20, 2023. The prior application Ser. No. 18/356,212 is a continuation application of and claims the priority benefit of a prior application Ser. No. 17/525,975, filed on Nov. 15, 2021. The prior application Ser. No. 17/525,975 is a divisional application of and claims the priority benefit of a prior application Ser. No. 16/547,605, filed on Aug. 22, 2019. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

[0002] Semiconductor devices are used in a variety of electronic applications, such as personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon. Many integrated circuits are typically manufactured on a single semiconductor wafer. The dies of the wafer may be processed and packaged at the wafer level, and various technologies have been developed for wafer level packaging.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 to FIG. 13 illustrate cross-sectional views of intermediate stages in the manufacturing of a semiconductor package according to some exemplary embodiments of the present disclosure.

DETAILED DESCRIPTION

[0005] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and

second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0006] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0007] Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

[0008] FIG. 1 to FIG. 13 illustrate cross-sectional views of intermediate stages in the manufacturing of a semiconductor package according to some exemplary embodiments of the present disclosure. It is noted that the present disclosure will be described with respect to some embodiments in a specific context, namely an integrated fan-out package structure. The concepts in the disclosure may also apply, however, to other semiconductor structures or circuits. A semiconductor package and the manufacturing method of forming the semiconductor package and semiconductor device applicable for the semiconductor package structure are provided in accordance with various embodiments. The intermediate stages of forming the semiconductor package are illustrated in accordance with some embodiments. The variations of the embodiments are discussed. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements.

[0009] With now reference to FIG. 1, in some embodiments, a semiconductor wafer **10** is provided. In accordance with some embodiments of the disclosure, the semiconductor wafer **10** may include a partially fabricated semiconductor substrate **112** having an active side **S1** and a back side **S2** opposite to the active side **S1**. Typically, wafer fabrication for semiconductor devices is almost finished at this stage and the partially fabricated substrate is to be packaged into a final product. For example, the substrate **112** may include many partially fabricated semiconductor devices, each in a separate region separated by scribe lines **L1**. Each of these separate regions becomes a die when the wafer is singularized, or diced, along the scribe lines **L1**.

[0010] In accordance with some embodiments of the disclosure, a plurality of conductive bumps **114** are formed on the active side **S1**. In some embodiments, the conductive bumps **114** may be arranged into various patterns or arrays and may be of size from several microns to hundreds of microns or larger. A perspective view of a substrate **112** having conductive bumps **114** formed thereon is shown as FIG. 1.

[0011] Referring to FIG. 1, the semiconductor wafer **10** is shown having the conductive bumps **114** formed thereon. The substrate **112** may include various embedded semiconductor metal and dielectric layers corresponding to circuitry and semiconductor devices, such as transistors, resistors, capacitors, and diodes (not shown). The circuitry and devices are electrically connected to the conductive bumps **114**. In some embodiments, the conductive bumps **114** may be formed by a plating or printing process and formed into a partial ball shape by heating the solder material.

[0012] With now reference to FIG. 2, a protection film **117** is laminated on the active side **S1** of the

semiconductor wafer **10**. In accordance with some embodiments of the disclosure, the protection film **117** may include a dielectric film **116** and a cover film **115**. In some embodiments, the dielectric film **116** covers the conductive bumps **114** and the cover film **115** covers the dielectric film **116** as shown in FIG. 2. For example, materials of the dielectric film **116** may include a B-stage epoxy resin and a filler. In general, B-stage is a stage that can be achieved by utilizing heat or UV light to remove the majority of solvent from an (epoxy) adhesive, thereby allowing construction to be staged. In between adhesive application, assembly, and curing, the B-stage epoxy resin can be held for a period of time, without sacrificing performance. In some embodiments, the dielectric film **116** may include reinforcing materials such as glass fabric, fiber paper, glass fiber paper, etc., which is soaked into resin (primarily epoxy resin), for example. In some embodiments, the dielectric film **116** contains more filler than a normal dielectric layer does. For example, a percentage of the filler in the dielectric film **116** is substantially from 50% to 80%. [0013] In accordance with some embodiments of the disclosure, the cover film **115** may be a release film, which provides protection to the dielectric film **116** and can be detached from the dielectric film **116** in the later process. In one of the implementations, the protection film **117** may be laminated on the active side **S1** by laminating the dielectric film **116** on the active side **S1** first, and then laminating the cover film **115** on the dielectric film **116**. That is to say, the dielectric film **116** and the cover film **115** can be laminated on the active side **S1** of the semiconductor wafer **10** sequentially. In an alternative embodiment, the dielectric film **116** and the cover film **115** can be bonded together and laminated on the active side **S1** simultaneously by a roller, for example. The disclosure is not limited thereto.

[0014] In accordance with some embodiments of the disclosure, after the protection film **117** is laminated on the active side **S1** of the semiconductor wafer **10**, a curing or baking process may be performed on the protection film **117** to cure the protection film **117**. In some embodiments, a temperature of the curing process is substantially below 200° C. to avoid melting of the conductive bumps **114**. In one of the implementations, the protection film **117** may be baked in a heating chamber at a temperature between 130° C. and about 200° C. In accordance with some embodiments of the disclosure, a plurality of nano-bubbles may be induced during the curing process. Accordingly, the cured dielectric film **116** may include a plurality of nano-bubbles. In some embodiments, a diameter of each of the nano-bubbles is from about 10 nm to about 100 nm.

[0015] With now reference to FIG. 3, a thinning process, which may be a grinding process, is performed on the back side **S2** of the semiconductor wafer **10** to form a thinned semiconductor wafer **10a**. The back side **S2** of the semiconductor wafer **10** may be ground to reduce the thickness thereof. With the mechanical grinding, the semiconductor wafer **10** can be ground to a thickness of about 50-100 μm , depending on the conditions. However, further reduction in the thickness of the semiconductor wafer **10** by mechanical grinding may result in damage to the wafer. Therefore, to further reduce the thickness of the semiconductor wafer **10**, it is preferable to use a method less likely to cause damage during a process, such as wet chemical etching or chemical mechanical polishing (CMP), but the disclosure is not limited thereto. It is noted that the thickness of the thinned semiconductor wafer **10a** can be set depending on the purpose for which the semiconductor package is used. During the thinning process, the semiconductor wafer **10** may be held by a handler from the side where the protection film **117** is attached. Hence the protection film **117** protects the conductive bumps **114** from the handler. The thinning process reduces the weight of the semiconductor devices formed from the thinned semiconductor wafer **10a** and such thinned semiconductor devices have important applications for some small portable devices.

[0016] In accordance with some embodiments of the disclosure, after the thinning process is performed on the back side **S2**, the thinned semiconductor wafer **10a** may then be flipped over and mounted on a frame carrier **20** with the active side **S1** facing up. In some embodiments, the frame carrier **20** may include a dicing frame **21** and a dicing tape **22**, but the disclosure is not limited thereto. In some embodiments, the dicing frame **21** may be a sheet metal frame in a ring shape, for

example, and the dicing tape **22** is adhered to the thinned semiconductor wafer **10a**.

[0017] With now reference to FIG. **4**, in some embodiments, the cover film **115** can be removed by a peeling process. In some embodiments, the cover film **115** may be directly peeled off from the dielectric film **116** without any further treatments. In an alternative embodiment, the cover film **115** may be peeled off after exposure of UV light or heat treatment for reducing adhesion of the cover film **115**. For example, the cover film **115** may be exposed to UV light, so that the cover film **115** loses or reduces adhesion, and hence the cover film **115** can be peeled off from the dielectric film **116**, but the disclosure is not limited thereto. After the cover film **115** is removed, the dielectric film **116** of the protection film **117** remains covering the conductive bumps **114**.

[0018] With now reference to FIG. **5**, in some embodiments, a singularization process is performed on the thinned semiconductor wafer **10a** with the dielectric film **116** on top (hereinafter “wafer **10a**”), so as to form a plurality of semiconductor devices **110a** as shown in FIG. **5**. In accordance with some embodiments of the disclosure, the wafer **10a** is then diced in a manner along the scribe lines **L1** to separate the wafer **10a** into individual semiconductor devices **110a**. In some embodiments, the wafer **10a** is sawed by a mechanical or laser blade along the scribe lines **L1** between adjacent semiconductor devices **110a**. In some embodiments, the blade is sawed through the wafer **10a** and into the dicing tape **22**, such that the semiconductor devices **110a** are separated from one another while still attached to the frame carrier **20** by the dicing tape **22** and maintained on the frame carrier **20**.

[0019] With now reference to FIG. **6**, in some embodiments, at least one of the plurality of semiconductor devices **110a** is picked up from the frame carrier **20** and is placed on another carrier **160**. In some embodiments, the carrier **160** may further include an adhesive layer **165**. The carrier **160** may be a glass carrier, a ceramic carrier, or the like. The adhesive layer **165** may be a light to heat conversion release coating (LTHC), or the like. In some embodiments, an insulation layer **170a** may be optionally provided on the carrier **160**, or on the adhesive layer **165** (if any). In some embodiments, before the semiconductor device **110a** is disposed on the carrier **160**, a plurality of conductive pillars **130** may be formed on the carrier **160**, and the conductive pillars **130** surrounds a device area **A1** of the carrier **160** where the semiconductor device **110a** is disposed. In the present embodiment, the conductive pillars **130** are formed on the insulation layer **170a** located on the carrier **160**, but the disclosure is not limited thereto.

[0020] In accordance with some embodiments of the disclosure, the semiconductor device **110a** formed by the process shown in FIG. **1** to FIG. **5** may include an active surface having a plurality of conductive bumps **114** and a dielectric film **116** encapsulating the conductive bumps **114**. In some embodiments, the materials of the dielectric film **116** may include an epoxy resin and a filler. In some embodiments, the dielectric film **116** may include reinforcing materials such as glass fabric, fiber paper, glass fiber paper, etc., which is distributed in the resin (primarily epoxy resin), for example. In some embodiments, the dielectric film **116** contains more filler than a normal dielectric layer does. For example, a percentage of the filler in the dielectric film **116** is substantially from 50% to 80%. In accordance with some embodiments of the disclosure, the dielectric film **116** may include a plurality of nano-bubbles, and the diameter of each of the nano-bubbles is from about 10 nm to about 100 nm, for example.

[0021] In some exemplary embodiments, the conductive bumps **114** (such as copper vias) may be formed on the active surface (e.g. the top surface) of the semiconductor device **110a** and electrically coupled to the bond pads **113** on the substrate **112** of the semiconductor device **110a**. The bond pads **113** may be partially covered by a passivation layer, which may be polyimide, a polymer adhesive, or other insulating buffer material. The conductive bumps **114** are disposed on the bond pads **113** and fill the opening between the passivation layer. In some embodiments, the dielectric film **116** covers the active surface (e.g. the top surface) of the semiconductor device **110a**, and may cover the top surfaces of the conductive bumps **114**. In other embodiments, the top surface of the dielectric film **116** may be substantially level with the top surfaces of the conductive bumps

114. In some embodiments, the top ends of the conductive pillars **130** may be substantially level with the top surfaces of the conductive bumps **114**. In other embodiments, the top ends of the conductive pillars **130** may be substantially higher than the top surfaces of the conductive bumps **114**. Alternatively, the top ends of the conductive pillars **130** may be substantially lower than the top surfaces of the conductive bumps **114** but substantially higher than the bottom surfaces of the conductive bumps **114**.

[0022] In accordance with some embodiments of the disclosure, the semiconductor device **110a** may be logic device dies including logic circuits therein. In some exemplary embodiments, the semiconductor device **110a** may be dies that are designed for mobile applications, and may include a Power Management Integrated Circuit (PMIC) die and a Transceiver (TRX) die, for example. Although one semiconductor device **110a** is illustrated, more dies may be placed over the carrier **160** and level with one another.

[0023] With such configuration, the process of laminating the protection film **117** on the active side **S1** of the semiconductor wafer **10** not only provides protection to the conductive bumps **114** during the thinning process, but also, after the cover film **115** is removed, the dielectric film **116** can function as a passivation layer for encapsulating the conductive bumps **114**. Accordingly, processes of attaching a backside grinding tape and forming a passivation layer on the active side **S1** of the semiconductor wafer **10** by spin coating, which is rather expensive and time consuming, can be omitted. Thereby, the manufacturing process of the semiconductor device **110a** and the semiconductor package **100** can be simplified, and material and production cost thereof can be saved. In addition, the curing temperature of the dielectric film **116** is rather low (e.g. below about 200° C.), so the issue of melting the conductive bumps **114** during the curing process can be avoided.

[0024] In some exemplary embodiments, the conductive pillars **130** may be pre-formed, and are then placed on the carrier **160**. In alternative embodiments, the conductive pillars **130** may be formed by, for example, plating process. The plating of the conductive pillars **130** may be performed before the placement of the semiconductor device **110a**, and may include forming a seed layer (not shown) over carrier **160**, forming and patterning a photo resist layer (not shown), and plating the conductive pillars **130** on the portions of the seed layer that are exposed through the photo resist layer. The photo resist layer and the portions of the seed layer covered by the photo resist layer may then be removed. The semiconductor device **110a** may then be placed over the carrier **160**. The material of the conductive pillars **130** may include copper, aluminum, or the like. Accordingly, the bottom ends of the conductive pillars **130** are substantially level with the back surface of the semiconductor device **110a**.

[0025] With now reference to FIG. 7, in some embodiments, the semiconductor device **110a** and the conductive pillars **130** on the carrier **160** are encapsulated by an encapsulating material **120**. In other words, the encapsulating material **120** is provided on the carrier **160** to encapsulate the conductive pillars **130** and the semiconductor device **110a** at the device area **A1**. In some embodiments, the encapsulating material **120** fills the gaps between the semiconductor device **110a** and the conductive pillars **130**, and may be in contact with the insulation layer **170a**. The encapsulating material **120** may include a molding compound, an epoxy, or a resin, etc. In some embodiments, a top surface of the encapsulating material **120** may be higher than the top ends of the conductive pillars **130** and the top surface of the dielectric layer **116a**. Namely, the encapsulating material **120** covers the top ends of the conductive pillars **130** and the top surface of the dielectric layer **116a**. In some embodiments, the encapsulating material **120** and the dielectric film **116** are two separate components, and the conductive bumps **114** are isolated from the encapsulating material **120** by the dielectric film **116**. In some embodiments, at least a part of the materials of the encapsulating material **120** and the dielectric film **116** are different from each other.

[0026] With now reference to FIG. 8, in some embodiments, a thinning process, which may be a grinding process, is performed to thin the encapsulating material **120** (and the dielectric film **116**)

until the top ends of the conductive pillars **130** and the top surfaces of the conductive bumps **114** are revealed. The resulting structure is shown in FIG. **8**. Due to the thinning process, the top ends of the conductive pillars **130** are substantially level with the top surfaces of the conductive bumps **114**, and are substantially level with the top surface of the encapsulating material **120** and the top surface of the dielectric film **116** as shown in FIG. **8**. In some embodiments, the thinning process may be implemented through mechanical grinding, wet chemical etching and/or chemical mechanical polishing (CMP), but the disclosure is not limited thereto.

[0027] Throughout the description, the resultant structure including the semiconductor device **110**, the conductive pillars **130** and the encapsulating material **120** as shown in FIG. **8** is referred to as encapsulated semiconductor device **101**, which may have a wafer form in the process. Accordingly, in the encapsulated semiconductor device **101**, the semiconductor device **110** is disposed at the die area **A1**, and the top surface of the dielectric film **116** may be substantially level with and reveal the top surfaces of the conductive bumps **114**. The conductive pillars **130** may extend through the encapsulated semiconductor device **101** outside of the die area **A1**, and the encapsulating material **120** encapsulates the semiconductor device **110** and the conductive pillars **130**. In other words, the encapsulating material **120** encapsulates the semiconductor device **110** therein, and the conductive pillars (through vias) **130** extends through the encapsulating material **120**.

[0028] With now reference to FIG. **9**, in some embodiments, a redistribution structure **140** is provided over the encapsulating material **120** and the semiconductor device **110**. The redistribution structure **140** is electrically connected to the semiconductor device **110** and the conductive pillars **130**. In some embodiments, the redistribution structure **140** is formed over the encapsulated semiconductor device **101** to be electrically connected to the conductive bumps **114** of the semiconductor device **110** and the conductive pillars **130**. In some embodiments, the redistribution structure **140** may also interconnect the conductive bumps **114** and the conductive pillars **130**. The redistribution structure **140** may be formed by, for example, depositing conductive layers, patterning the conductive layers to form redistribution circuits **142**, partially covering the redistribution circuits **142** and filling the gaps between the redistribution circuits **142** with dielectric layers **143**, etc. The material of the redistribution circuits **142** may include a metal or a metal alloy including aluminum, copper, tungsten, and/or alloys thereof. The dielectric layers **143** may be formed of dielectric materials such as oxides, nitrides, carbides, carbon nitrides, combinations thereof, and/or multi-layers thereof. The redistribution circuits **142** are formed in the dielectric layers **143** and electrically connected to the semiconductor device **110** and the conductive pillars **130**. In addition, an under bump metallurgy (UBM) layer **144** may be formed on the redistribution structure **140** by sputtering, evaporation, or electroless plating, etc.

[0029] With now reference to FIG. **10**, in some embodiments, at least one of electrical connector **182** and at least one integrated passive device (IPD) **184** are disposed on the redistribution structure **140** in accordance with some exemplary embodiments. The formation of the electrical connectors **182** may include placing solder balls on the UBM layer **144** (or on the redistribution structure **140**), and then reflowing the solder balls. In alternative embodiments, the formation of the electrical connectors **182** may include performing a plating process to form solder regions on the UBM layer **144** (or on the redistribution structure **140**), and then reflowing the solder regions. The electrical connector **182** may also include conductive pillars, or conductive pillars with solder caps, which may also be formed through plating. The IPD **184** may be fabricated using standard wafer fabrication technologies such as thin film and photolithography processing, and may be mounted on the redistribution structure **140** through, for example, flip-chip bonding or wire bonding, etc.

[0030] With now reference to FIG. **11**, in some embodiments, the carrier **160** may be removed. In some embodiments, the carrier **160** is detached from the encapsulated semiconductor device **101**, and the insulation layer **170a** (if any), by causing the adhesive layer **165** to lose or reduce adhesion. The adhesive layer **165** is then removed along with the carrier **160**. For example, the adhesive layer **165** may be exposed to UV light, so that the adhesive layer **165** loses or reduces adhesion, and

hence the carrier **160** and the adhesive layer **165** can be removed from the encapsulated semiconductor device **101**.

[0031] With now reference to FIG. **11**, in some embodiments, after the carrier **160** is removed, the bottom ends of the conductive pillars **130** are revealed. In the illustrated structure, the bottom ends of the conductive pillars **130** are level with the bottom surface of the semiconductor device **110** and the bottom surface of the encapsulating material **120**. In the embodiments of the insulation layer **170a** being omitted, a grinding process may be performed to lightly grind the back surfaces of the semiconductor device **110** and the encapsulating material **120**, and the bottom ends of the conductive pillars **130**. Alternatively, the grinding process may be skipped.

[0032] In the embodiments having the insulation layer **170a**, a patterning process may then be performed on the insulation layer **170a** to form a plurality of openings **172**. Accordingly, the insulation layer **170** having a plurality of openings **172** are formed. The openings **172** may be located on the conductive pillars **130** respectively to reveal the bottom ends of the conductive pillars **130**. In some embodiments, the openings **172** may be formed by photolithography process, laser drilling process, etc. Then, a plurality of electrical terminals **190** may be formed on the encapsulated semiconductor device **101** to be electrically connected to the conductive pillars **130**. In some embodiments, the electrical terminals **190** are disposed in the openings **172** of the insulation layer **170** to be connected to the conductive pillars **130**. At the time, the manufacturing process of a semiconductor package **100** may be substantially done.

[0033] With now reference to FIG. **11**, in some embodiments, a second package **200** may be mounted on and electrically connected to the first package **100** through the electrical terminals **190**. Accordingly, the resulting structure as shown in FIG. **13** is a package on package structure. In some embodiments, the second package **200** may be packages, device dies, passive devices, and/or the like. In some embodiments, the package on package structure shown in FIG. **13** may combine vertically discrete memory and logic packages, and the second package **200** may be employed in a memory such as a dynamic random access memory (DRAM) and others, but the disclosure is not limited thereto.

[0034] In some embodiments, the bonding between the semiconductor package (first package) **100** and the second package **200** may be performed using flip-chip bonding through the electrical terminals **190**, which may comprise solder, for example. In some embodiments, an underfill **240** may be formed between the first package **100** and the second package **200** to encapsulate the electrical terminals **190**. It is appreciated that the semiconductor device **110** in the first package **100** and the semiconductor device **220** in the second package **200** may be arranged differently than in the illustrated exemplary embodiments. In some embodiments, the semiconductor device **220** is encapsulated by the insulating encapsulation **230**. Then, the wafer-level package may then be sawed into a plurality of package on package structures independent from one another, with each of the package on package structures including one second package **200** bonded to one first package **100**.

[0035] In sum, the protection film **117** including the dielectric film **116** and the cover film **115** covering the dielectric film **116** is laminated on the active side **S1** of the semiconductor wafer **10**. With such configuration, the protection film **117** not only provides protection to the conductive bumps **114** on the semiconductor wafer **10** during the thinning process, but also, after the cover film **115** is removed, the dielectric film **116** can function as a passivation layer for encapsulating the conductive bumps **114**. Accordingly, processes such as attaching a backside grinding tape and forming a passivation layer on the active side **S1** of the semiconductor wafer **10** by spin coating, which is rather expensive and time-consuming, can be omitted. Thereby, the manufacturing process of the semiconductor device and the semiconductor package can be simplified, and material and production cost thereof can be saved. In addition, the curing temperature of the dielectric film **116** is rather low (e.g. below about 200° C.), so the issue of melting the conductive bumps **114** during the curing process can be avoided.

[0036] Based on the above discussions, it can be seen that the present disclosure offers various advantages. It is understood, however, that not all advantages are necessarily discussed herein, and other embodiments may offer different advantages, and that no particular advantage is required for all embodiments.

[0037] In accordance with some embodiments of the disclosure, a semiconductor package includes a semiconductor device, an encapsulating material encapsulating the semiconductor device, and a redistribution structure disposed over the encapsulating material and the semiconductor device. The semiconductor device includes an active surface having conductive bumps and a dielectric film encapsulating the conductive bumps, where a material of the dielectric film comprises an epoxy resin and a filler. The conductive bumps are isolated from the encapsulating material by the dielectric film, and the redistribution structure is electrically connected to the conductive bumps.

[0038] In accordance with some embodiments of the disclosure, a semiconductor package includes a semiconductor device, an encapsulating material extending along sidewalls of the semiconductor device, and a redistribution structure. The semiconductor device includes a semiconductor substrate, conductive bumps disposed over the semiconductor substrate, and a laminated film disposed over the semiconductor substrate and extending along sidewalls of the conductive bumps. The redistribution structure overlies a top surface of the encapsulating material and top surfaces of the laminated film and the conductive bumps.

[0039] In accordance with some embodiments of the disclosure, a manufacturing method of a semiconductor package includes the following steps. A dielectric film is laminated over a semiconductor substrate to cover conductive bumps that are formed over an active side of the semiconductor substrate. A backside thinning process is performed to reduce a thickness of the semiconductor substrate, where during the backside thinning process, the dielectric film protects the conductive bumps. The dielectric film and the semiconductor substrate are cut through to form a semiconductor device. An encapsulating material is formed to laterally cover the semiconductor device, where top surfaces of the dielectric film and the conductive bumps are substantially leveled with a top surface of the encapsulating material. A redistribution structure is formed on the encapsulating material and the semiconductor device.

[0040] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor package, comprising: a semiconductor device comprising a semiconductor substrate, conductive bumps over the semiconductor substrate, a dielectric film encapsulating the conductive bumps, and a passivation layer disposed between the semiconductor substrate and the dielectric film, wherein a material of the dielectric film comprises an epoxy resin and a filler, an outer sidewall of the dielectric film and an outer sidewall of the semiconductor substrate are contiguous to each other, and the dielectric film is in contact with the passivation layer; and an encapsulating material encapsulating the semiconductor device.
2. The semiconductor package as claimed in claim 1, further comprising: a redistribution structure disposed over the encapsulating material and the semiconductor device, wherein a conductive material of the redistribution structure is electrically connected to the conductive bumps, and a dielectric material of the redistribution structure is in direct contact with the dielectric film.

3. The semiconductor package as claimed in claim 1, further comprising: a conductive pillar penetrating through the encapsulating material; and an electrical terminal electrically coupled to the conductive bumps of the semiconductor device through the redistribution structure and the conductive pillar, and the electrical terminal and the redistribution structure being disposed at two opposing sides of the conductive pillar.
4. The semiconductor package as claimed in claim 1, wherein an inner sidewall of the encapsulating material is in direct contact with the outer sidewall of the dielectric film and the outer sidewall of the semiconductor substrate.
5. The semiconductor package as claimed in claim 1, wherein the outer sidewall of the dielectric film and the outer sidewall of the semiconductor substrate are substantially flush with each other.
6. The semiconductor package as claimed in claim 1, wherein the dielectric film comprises nano-bubbles.
7. The semiconductor package as claimed in claim 1, wherein a top surface of the dielectric film is leveled with top surfaces of the conductive bumps.
8. The semiconductor package as claimed in claim 1, wherein an inner sidewall of the dielectric film is in direct contact with sidewalls of the conductive bumps.
9. A manufacturing method of a semiconductor package, comprising: forming a plurality of conductive connectors on a first side of a semiconductor substrate, laminating a dielectric material on the semiconductor substrate, the dielectric material covering and surrounding the conductive connectors; thinning the semiconductor substrate from a second side opposite to the first side; and dicing the semiconductor substrate with the dielectric material to form at least one semiconductor device after the thinning, wherein after the dicing, the conductive connectors are buried in the dielectric material.
10. The manufacturing method as claimed in claim 9, wherein the dielectric material is a curable material.
11. The manufacturing method as claimed in claim 10, wherein the curable material comprises B-stage epoxy and fillers.
12. The manufacturing method as claimed in claim 11, wherein a percentage of the fillers in the curable material is from 50% to 80%.
13. The manufacturing method as claimed in claim 9, further comprising laminating a cover material on the dielectric material.
14. The manufacturing method as claimed in claim 13, further comprising curing the dielectric material after laminating the cover material and before thinning the semiconductor substrate.
15. The manufacturing method as claimed in claim 9, wherein the dielectric material comprises a plurality of nano-bubbles.
16. The manufacturing method as claimed in claim 9, further comprising: forming an encapsulating material to surround a sidewall of the at least one semiconductor device, wherein the top surfaces of the conductive connectors are separated from the encapsulating material by the dielectric material; and performing a planarization process on the encapsulating material, wherein after the planarization process, planarized surfaces of the encapsulating material, the dielectric material, and the conductive connectors are substantially leveled with one another.
17. A manufacturing method of a semiconductor package, comprising: providing a wafer having a plurality of device regions separated by scribed lines; forming a plurality of protruding features on a first side of the wafer in each of the device regions; forming a protection film on the wafer, the protection film covering the protruding features in each of the device regions and covering the scribe lines; thinning the wafer from a second side opposite to the first side; and removing a portion of the protection film after the thinning, wherein the protruding features are buried in the remaining protection film.
18. The manufacturing method as claimed in claim 17, further comprising: performing a singularization process along the scribe lines to separate the device regions with the remaining

protection film and therefore form a plurality of semiconductor devices.

19. The manufacturing method as claimed in claim 17, further comprising: removing another portion of the protection film until the protruding features in each of the device regions are exposed.

20. The manufacturing method as claimed in claim 17, wherein the protection film has a multi-layer structure.
